Statistics and Trends: Analysis of Fake and Real News

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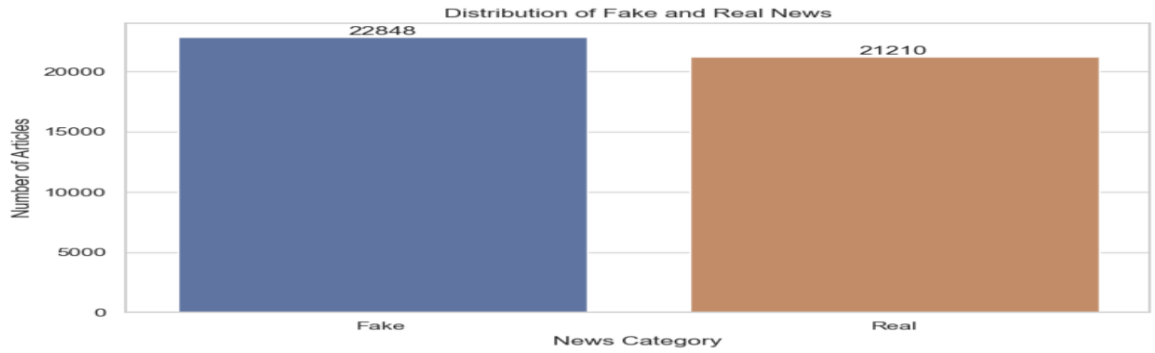
1. Introduction

In this report I will examine a set of news articles that are classified as Fake or Real. I was interested in the number of articles in each group, if there was any relationship between the length of a headline and the length of the article, and whether there was any difference in the length of the article between the two groups. I used the two supplied files, Fake.csv and True.csv. I merged, added a category label, deleted duplicate data and excluded rows with missing, or empty, article text or title. Following these operations 44,058 records of books, journals, and other materials remained.

For every record, I recorded the number of words in an article, the number of words in an article's title, and the number of characters in the text of the article. These measures enabled me to compare the size of the articles in the same manner over the entire data set. The figures and statistics listed below are for cleaned records and should not be considered as a confirmation of whether a particular news story is true or false.

2. Category distribution

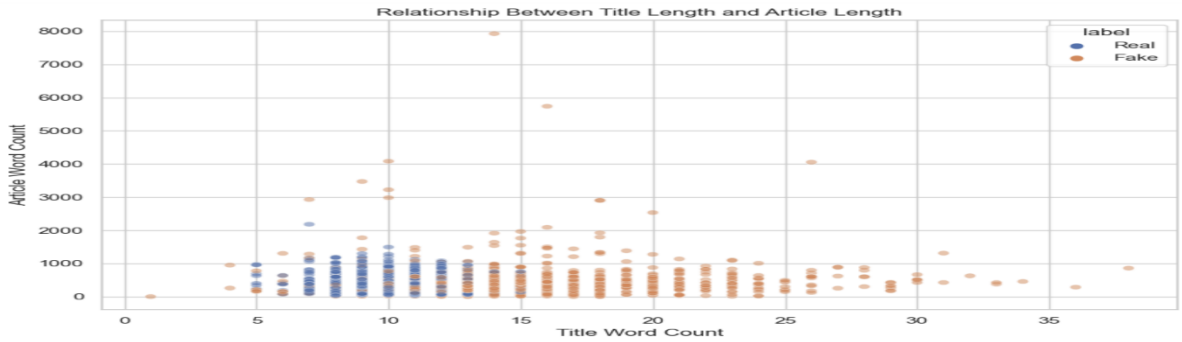
51.87% of the cleaned data is Fake articles (22,848 items). The remaining 21,210 articles are labelled Real, or 48.13%. There's not that much difference between the groups, with Fake having 1638 more records. This implies that there is a reasonably even distribution of the data with respect to the comparison and not just one category. This difference can clearly be seen in Figure 1. The percentages are useful in conjunction with the counts, as it indicates the proportion of the entire data set.



If you compare the length of the articles in both groups, the average is 410.77 words and the median (middle article) is 365 words. The standard deviation of the articles is 350.70 words, indicating that lengths are quite different. The average is above the median, which indicates that there are fewer, but longer, articles that are affecting the average. The average number of words is shorter: 12.43, median 11. On average, articles have 2,501.60 characters, whereas the median has 2,206. These measures show the same trend with a higher mean than median.

3. Relationship between title length and article length

In Graphs, the number of words in a title is compared to the number of words in its article. For readability, a subset of 2,500 records is shown in the graph. The overall Pearson correlation is 0.057. This value is very close to zero, indicating little linear relationship between the two measurements in this data set. If the categories are considered separately, the figures are also very small: 0.034 for Fake articles and −0.038 for Real articles.

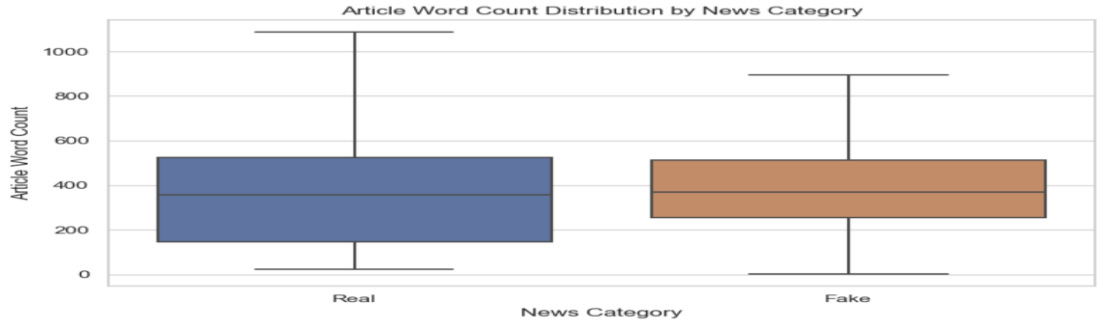


No distinct down or up trend of the points. A short headline doesn't always mean a short article and vice versa. On the basis of these findings, the length of headlines does not provide a good indicator of article length. The correlation only describes a linear relation, it excludes all other potential relations.

4. Article length by category

On average a Fake article has 434.89 words compared with 384.78 words for a Real article. Their medians are a lot closer, though: 369 words for Fake and 359 words for Real. It is important to note this difference when reading the graph. Although the higher average article length is in the Fake group, the typical article in each group is fairly similar. The average seems to be more influenced by very long articles.

The difference is also larger for Fake articles. For Real articles, a standard deviation of 407.82 words is common, whereas it is 273.90 for their articles. The longest Fake article is 8,135 words long, the longest Real article is 5,172 words long. The middle regions of these two distributions are given in Figure 3. The Fake group has a bigger range, but the medians are not very different. Max values are shown in the text, while the maximum values of the outliers are not shown in the figure for clarity in comparison of the main distributions.



5. Conclusion

This comparison shows that there are approximately the same number of records in each of the two categories and that the median article lengths are also around the same length. The average length of the fake articles is longer and has a greater range, indicating that there is greater influence of unusually long articles on this group's average. There was little to no linear relationship between the length of the title and the length of the article. These results are for the "cleaned" dataset used in this analysis. The pattern may be altered if the original collection is incomplete or records are repeated, as removing those records will also impact the pattern. More data would be required to make generalizations about news articles.

Git Hub Url - https://github.com/SanthoshKumarGatla/Statistics-and-Trends